MILITARY SPECIFICATION

CYLINDER, HIGH PRESSURE AIR, FIBERGLASS REINFORCED PLASTIC (FRP), METAL LINED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is applicable for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fiberglass filament reinforces plastic cylinders having metal liners.

1.2 Classification. Cylinders snail be of the following types as specified (see 6. 1 and 6.2.1):

Type I - 250 cubic inch (in^3) volume.
Type II - 455 in^3 volume.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on sate of invitation for bids or request for proposal, form a part of this specification to the extent specifies herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Woos, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Snipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard. Corrugated, Triple-Wall.

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MIL-P-116 - Preservation-Packag ing, Methods of.
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap: Water-Waporproof or Waterproof, Flexible.
MIL-P-17549 - Plastic Laminates, Fibrous Glass Reinforce, Marine Structural.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-R-21607 - Resins, Polyester, Low-Pressure Laminating, Fire-Retardant.
MIL-I-22110 - Inhibitors, Corrosion, Volatile, Crystalline.
MIL-P-23377 - Primer Coating; Epoxy Polyamide, Chemical and Solvent Resistant.
MIL-P-24441/2 - Paint, Epoxy-Polyamide, Exterior Topcoat, Haze Gray, Formula 151.
MIL-C-47257 - Compound, Epoxy, Filament Winding.
MIL-R-60346 - Roving, Glass, Fibrous (For Filament Winding Applications).

STANDARDS

FEDERAL

FED-STD-406 - Plastic, Methods of Testing.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MS26545 - Cylinder, Compressed Gas, Nonshatterable.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

Computer Code NASA CR-72124 - Computer Program for the Analysis of Filament Wound Reinforced Metal Shell Pressure Vessels - May, 1966.

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(Application for copies should be addressed to the National Aeronautics and Space Administration, Washington, DC 20546.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article inspection. Prior to beginning production, sample cylinders snail be examined and tested as specified in 4.3 (see 6.3). Ability *of* the cylinders to meet the performance and environmental requirements of this specification shall be demonstrated by the first article inspection. Materials, design, or construction shall not be altered following first article acceptance.

3.2 Design and construction.

3.2.1 Design. The cylinders shall consist of fiberglass filaments impregnated with a thermosetting plastic resin and wound on a metal liner. The metal liner shall not be stressed above its elastic limit at cylinder pneumatic pressure up to proof pressure: the liner shall not be stressed above 95 percent of its compressive yield strength, cylinder unpressurized; and the maximum strength in the glass filaments at the 6300 pounds per square inch (lb/in^2), 160°F operating pressure shall not exceed 30 percent of the filament stress at virgin hydrostatic burst pressure of the cylinder. Stresses shall be computed by the Computer Code NASA CR-72124. or other equivalent analytic techniques.

3.2.1.1 Operating characteristics . Operating characteristics of the cylinder shall be as follows:

(a) Operating pressure - Normal room temperature (75°F ± 5°F) - 5000 lb/in^2
- High temperature (160°F) - 6300 lb/in^2
Low temperature (minus 20°F) - 3700 lb/in^2
(b) Proof pressure - 7500 lb/in^2 (minimum)
(c) Burst pressure - 15,000 lb/in^2 (minimum)
(d) Temperature range - Minus 20°F to 160°F
(e) Useful cyclic life (without liner leakage) - 500 fill cycles or 100 fill cycles plus 3500 temperature cycles from minus 20°F to 160°F
(f) Dynamic fatigue life - 10 years
(g) Leakage less than 5 percent pressure in *10* years

3.2.1.2 Physical characteristics. Physical characteristics shall be as follows:

(a) Port fitting - 302, 304 or 316 series CRES 1-1/8 inch 12-UNF-3A external thread (port fitting detail shall be in accordance with detail "X" of MS26545 for internal and external dimensions).

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		Type I	Type 11
(b)	Outside diameter	5.75 inches maximum	6.75 inches maximum
(c)	Length (overall)	20.5 inches maximum	23.5 inches maximum
(d)	Weight (overall)	14 lb maximum	20 lb maximum
(e)	Internal volume	250 in^3 $\pm$ 5 in^3	455 in^3 + 10 or -5 in^3

(f) Shape - Cylinder envelope shall be cylindrical in accordance with MS26545.

3.2.2 **Materials.** The cylinder shall be constructed of a liner of aluminum alloy see 3.2.7) reinforced with continuous filament glass fibers in accordance with type III or IV, class I of MIL-R-60346 and epoxy or polyester resin in accordance with MIL-C-47257 or MIL-R-21607, respectively. A resin system composed of Dow Chemical DER 332 and curing agent 'Y", or equal, may be used.

3.2.2.1 Material deterioration. Materials shall not deteriorate when subjected to the environmental conditions specified herein.

3.2.2.2 Liner. Metal liner shall be aluminum alloy UNS A96061, O temper, neat treated to T6 condition after final forming. Other metals or alloys may be proposed to the command or agency concerned. Metal liner material, design, and fabrication process shall not be altered from that demonstrated to have acceptable life by first article inspection. Liner alloy and fabrication process snail be identified on the detail drawings, and in the first article inspection report (see 4.3.2.1). No fissure or other defect is acceptable that is likely to weaken the finished cylinder. Smooth and uniform surface finish is required, no interior folding in the neck area of the liner is permitted; smooth gathering of the material in the neck is acceptable.

3.2.2.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.3 Process details. The following process details shall be followed in the fabrication of the cylinders:

(a) Guide rolls shall be not less than 2 inches in diameter for dry roving and not less than 1-1/2 inches in diameter for resin impregnated roving to avoid unnecessary chafing and flexing of the glass filaments.

(b) Under no circumstances shall dry roving be permitted to come into sliding contact with any elements of the winding machine, with the exception of the 2 inch diameter guide bar at the entrance of the resin pan.

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(c) Wetting time shall be not less than 0.5 second, if the wet-winding process is used.
(d) A Prony-brake device or an eddy-current brake device or equivalent device shall be used to maintain the proper tension in the glass roving during winding.

3.2.4 Process controls. A rigid process control shall be established and maintained to ensure compliance with the following conditions:

(a) Completely impregnated roving.
(b) Average resin content of the full wall thickness at any point, between 17 and 30 percent. Resin content of not less than 15 percent in any sample (see 4.6.12).
(c) Filament tension schedule such that the Inner plies of the glass filaments (roving) are not deformed out of their initial configuration by succeeding plies of roving.

3.2.5 Void content. Void content shall not exceed an average of 7 percent for all determinations on any cylinder when tested as specified in 4.6.13. Void content for any single determination shall not exceed 8 percent.

3.2.6 Protective coatings.

3.2.6.1 Metal coating and assembly. All surfaces of the liner not covered with fiberglass laminate shall be coated with MIL-C-5541 chemical treatment. After treatment is air dry, the port fitting shall be assembled into the cylinder opening with ‘On-ring seal equal to C. E. Conover and Company part number 3-912, and tightened to a torque of 360 to 420 inch-pounds (in-lb). Within 10 hours of application of the MIL-C-5541 treatment, exterior metal surfaces (not inside screw threads) on the blank end of the cylinder and on the port end, including the port fitting up to approximately 1/16 inch onto the wrench flats, shall be coated with MIL-P-23377 material to a dry film thickness of 0.5 mil to 0.75 roil, (if 10 hours is exceeded the surfaces shall be solvent cleaned immediately prior to the MIL-P-23377 application). Within 24 hours of application of MIL-P-23377 material, the coated metal surfaces shall be given two coats of formula 151 (gray) of MIL-P-24441/2. Each coat shall be 6 to 8 roils dry film thickness. At the port end, the objective is to coat exposed aluminum and to continue the coating, as a heavy fillet onto the port fitting, but leaving the major area of the wrench flats and outer end unpainted. Drying time between coats shall be no less than 6 hours or until a "tack free” state is achieved.

3.2.6.2 Barrier coating. A transparent moisture-barrier coating shall be applied to the exterior surface of the cylinder. The coating shall be sufficiently flexible to resist crazing, cracking. blistering or separation during pressure and environmental testing required herein.

3.2.7 Volume (for internal volumes, see 3.2.1.2). Volume shall be determined after the volumetric expansion test (see 4.6.1).

3.2.8 Structural strength. The cylinder shall have ample strength and rigidity to withstand 250 percent of maximum wrench torques as . specified in 3.2.6.1 without permanent deformation of the cylinder.

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3.3 Performance.

3.3.1 Volumetric expansion. First article inspection and production cylinders shall not have residual expansion greater than 10 percent of the elastic expansion when tested as specified in 4.6.1. The first 25 production cylinders shall not have an elastic expansion that exceeds by more than 10 percent the maximum acceptable value determined in the first article Inspection. The standard deviation shall be calculated for elastic expansion for the first article cylinders and for the first 25 production cylinders as a combined sample. Subsequent production cylinders shall not exceed the average elastic expansion for the combined sample by more than three times the standard deviation. Elastic expansion is the total expansion less the residual expansion.

3.3.2 Leakage. The rate of leakage of the cylinder shall not exceed one cubic centimeter (cm^3) per hour at any temperature within the design temperature range when tested as specified in 4.6.3.

3*3.3 Endurance. The cylinders, when tested as specified in 4.6.6, snail show no deterioration. The first cylinder shall meet 3.3.2 and shall be capable of withstanding a hydrostatic burst pressure of not less than 2.2 times room temperature operating pressure. The second cylinder shall withstand a hydrostatic burst pressure of not less than 9450 lb/in^2 (1.5 times 160°F, operating pressure of 6300 lb/in^2).

3.3.4 Thermal-shock and pneumatic leakage. The cylinders subjected to the thermal shock test of 4 6 5 shall show no deterioration and snail subsequently pass the pneumatic leakage tests (see 4.6.3.1 and 4.6.3.2).

3.3.5 Vibration and pneumatic leakage. The cylinder shall meet the vibration requirements of MIL-STD-167-1, type I. The cylinder shall show no deterioration upon completion of the test in 4.6.4.

3.3.6 Fragmentation resistance. The cylinder shall be resistant to fragmentation when penetrated by a projectile. The cylinder, when tested as specified in 4.6.8, shall remain in one piece, and the greatest dimension of the opening (cut plus tear) created by the projectile shall not exceed the dimension of one hole (cut) created by the projectile by more than 3 inches in any direction. “Cutting” snail be considered as the actual section of the cylinder cut by contact with the projectile, and a “tear” shall be considered as any extension beyond the cut.

3.3.7 Soak cycle. After subjection to the soak cycle test (see 4.6.9), the cylinder shall withstand a hydrostatic burst pressure of not less than 2.2 times room temperature operating pressure.

3.3.8 Pressure storage. The cylinder subjected to the pressure storage test (see 4. 6.10) shall withstand a hydrostatic burst pressure of not less than 2.2 times room temperature operating pressure.

3.3.9 Humidity cycle. The cylinder subjected to the humidity cycle test (see 4.6. 11) shall subsequently pass the hydrostatic burst test at a minimum pressure of 2.2 times the room temperature operating pressure.

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3.3.10 Pressure cycle. The cylinder shall pass the leakage test (see 3.3.2) after being subjected to 25 cycles. zero to opetating pneumati (air) pressure.

3.3.11 Virgin hydrostatic burst. The cylinder shall withstand a hydrostatic burst test (see 4.6.7) pressure of not less than 15,000 lb/in and when the pressure is increased to failure, the failure shall occur in the cylindrical portion of the vessel wall.

3.3.12 Impact resistance. After being subjected to the drop test (see 4.6.14), the cylinder shall pass the extreme temperature leakage test (see 4.6.3.2), the thermal shock (see 4.6.5) and pressure storage (see *4.6.10)* tests and shall show no signs *of* permanent deformation or structural damage.

3.4 Marking. The following shall be legibly and permanently marked in red letters on the cylinder in two places, 180 degrees apart:

"CHARGE WITH BREATHABLE COMPRESSED AIR
ONLY 5000 LB/IN^2 OPERATING PRESSURE AT
70°F HANDLE WITH CAUTION DO NOT DROP"

Letters shall be not less than 3/8-inch high.

3*5 Identification of product. The cylinders shall be marked for identification in accordance with MIL-STD-130, Specifying the following information only:

(a) Cylinder, high pressure air.
(b) Number of this specification.
(c) National Stock Number.
(d) Manufacturer's serial number and part number.
(e) Manufacturer's name or trademark (no address) and manufacturer's code identification number.
(f) Type .
(g) Date *of* manufacture (month and year).
(h) Date *of* test (month and year).

3.6 Workmanship Workmanship shall be of the quality necessary to produce cylinders free from defects which would affect proper functioning in service.

3.6.1 Surfaces. Interior and exterior surfaces of the cylinders shall be clean and free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specifie in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.3.1.

4.3.1 Samples for first article inspection. Six cylinders of each size and shape shall be submitted for the tests specified in 4.3.2. Each sample snail be identified with the manufacturer's serial number and as specified by the contracting activity (see 6.2.1).

4.3.2 First article inspection. First article inspection snail be conducted in accordance with the following schedule. Approval by similarity to previous inspection may be possible where such documentation is acceptable to the contracting activity. Factors to be considered include similarity of cylinder design, operating characteristics, materials, production history, and fabrication technique, in addition to actual inspection data. Examination and tests snail be conducted in the order listed:

(a) One cylinder shall be examined and tested as follows:
  (1) Volumetric expansion test (see 4.6.1).
  (2) Examination (see 4.6.2).
  (3) Pressure cycle test (see 4.6.15).
  (4) Extreme temperature leakage tests (see 4.6.3.2).
  (5) Fragmentation test (see 4.6.8).
  (6) Resin content determination (see 4.6.12).
(b) Two cylinders snail be examined and tested as follows:
  (1) Volumetric expansion test (see 4.6.1).
  (2) Examination (see 4.6.2).
  (3) Extreme temperature leakage tests (see 4.6.3.2).
  (4) Endurance test (see 4.6.6).
(c) One cylinder shall be examined and tested as follows:
  (1) Volumetric expansion test (see 4,6.1).
  (2) Examination (see 4.6.2).
  (3) Extreme temperature leakage tests (see 4.6.3.2).
  (4) Soak cycle test (see 4.6.9).
  (5) Resin content determination (see 4.6.12).
  (6) Void content (see 4.6.13).
(d) One cylinder shall be examined and tested as follows:
  (1) Volumetric expansion test (see 4.6.1).
  (2) Examination (see 4.6.2).
  (3) Virgin hydrostatic burst (see 4.6.7).
(e) One cylinder snail be examined and tested as follows:
  (1) Volumetric expansion test (see 4.6.1).
  (2) Examination (see 4.6.2).
  (3) Extreme temperature leakage test (see 4.6.3.2).
  (4) Vibration test (see 4.6.4).
  (5) Drop test (see 4.6.14).
  (6) Examination (see 4.6.2).
  (7) Extreme temperature leakage test (see 4.6.3.2).
  (8) Thermal-shock test (see 4.6.5).

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(9) Pneumatic and extreme temperature leakage test (see 4.6.3.1 and 4.6.3.2).
(10) pressure storage test (see 4.6.10).

4.3.2.1 First article test report. The contractor shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following tests:

(a) Individual tests (see 4.4.2).
(b) Lot sampling tests (see 4.4.3).

4.4.1 Lot. A lot shall consist of 200 cylinders or any lesser number offered for delivery at one time.

4.4.2 Individual tests. Cylinder shall be subjected to the following examination and tests in the order listed:

(a) Volumetric expansion test (see 4.6. 1).
(b) Leakage test (see 4.6.3.1).
(c) Examination (see 4.6.2).

Note: The empty weight may be taken during the volumetric expansion test and the volume determined after that test and prior to the leakage test.

4.4.3 Sampling. Two cylinders snail be selected at random from each lot offered for delivery. Assemblies which have been determined discrepant for characteristics other than those which would affect testing may be used as sample units.

4.4.3.1 Tests shall be conducted in the order listed:

(a) One cylinder shall be subjected to the following examinatio and tests:
(1) Volumetric expansion test (see 4.6.1).
(2) Examination (see 4.6.2).
(3) Leakage test (see 4.6.3.1).
(4) Virgin hydrostatic burst test (see 4.6.7).
(5) Resin content determination (see 4.6.12).
(6) Void content (see 4.6.13).
(b) One cylinder shall be subjected to the following examinatio and tests:
(1) Volumetric expansion test (see 4.6.1).
(2) Examination (see 4.6.2).
(3) Leakage test (see 4.6.3.1).
(4) Humidity cycle test (see 4.6.11).
(5) Resin content determination (see 4.6.12).
(6) Void content (see 4.6.13).

4.4.4 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand from that lot or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall

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inform the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.4.4.1 Individual tests may continue. Individual tests and production may be continued pending the investigation *of* a sample test failure, however, final acceptance of items on hand or produced later, shall not be made until it is determined that all items meet all the requirements.

4.4.4.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already delivered or in service. If the type of defect found denotes an operational safety hazard, the contracting activity shall be notified at once. At the conclusion of the investigation, the producer shall fully advise the contracting activity of all the defects found and the method of correcting them.

4.5 Test conditions.

4.5.1 Pressure media. Pressure tests shall be performed pneumatically with air or a helium-air mixture, except that volumetric expansion, endurance, hydrostatic burst, soak cycle, and humidity cycle tests may be performed with fresh water. The permissible tolerance snail be plus or minus 10 lb/in^2 from the stated value.

4.5.2 Temperature. Unless otherwise specified herein, tests shall be conducted at room temperature, that is between 70°F and 80°F. The temperature of the gas supply shall also be maintained within these limits for room temperature tests. The actual ambient temperatures shall be recorded . Permissible tolerances shall be +10°F for minus 20°F, +5°F for 160°F, and $\pm$5°F for 120°F.

4.5.3 Rate of pressure rise. The rate of pressure rise for the volumetric expansion test and the hydrostatic burst test shall be not greater than 25,000 lb/in^2 per minute.

4.5.4 Standard air. Standard air shall be considered to be dry air at a temperature of 69°F, pressure of 14.7 lb/in^2, and weighing 0.075 pounds per cubic foot (density).

4.6 Examination and tests.

4.6.1 Volumetric expansion test. The volumetric expansion test shall be conducted by subjecting the cylinder to 1.5 times operating pressure and maintaining that pressure for not less than 60 seconds to insure complete expansion of the liner. The test shall be performed individually and the volumetric expansion of the cylinder shall be determined to 0.5 milliliters (mL) or to an accuracy of 1 percent of total expansion and recorded in mL. No pressure in excess of operating pressure shall be applied prior to this test unless It is a pre-stressing operation as part of the manufacturing process. The total l xpansion shall be recorded and the cylinder shall be repressurized and the residual expansion shall be recorded. Cylinder test results shall meet 3.3.1. The rate of pressure rise shall be in accordance with 4.5.3.

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4.6.2 Examination. The cylinder shall be carefully examined to determine conformance to the requirements of this specification including design, volume, weight, workmanship, marking, and for any visible defects.

4.6.3 Leakage tests.

4.6.3.1 Pneumatic leakage test. The cylinder shall be subjected to pneumatic leakage test at normal operating pressure and ambient temperature. Leakage shall not exceed the rate allowable in 3.3.2. The method used to determine the rate of leakage shall be reported and shall be capable of measuring leakage of 1 x 10^{-4} cm^3 per second. Leakage shall be measured over a period of not less than 5 minutes.

4.6.3.2 Extreme temperature leakage test. The cylinder shall be subjected to a pneumatic leakage test at a pressure of 3700 lb/in^2 and at a temperature of minus 20°F. After the temperature of the cylinder has stabilized, the leakage rate shall not exceed the permissible rate stated in 3.3.2. Test shall be repeated at a temperature of 160°F and at a pressure of 6300 lb/in^2. After the temperature of the cylinder has stabilized, the leakage rate shall not exceed the rate permissible In 3.3.2. Leakage shall be measured over a period of not less than 15 minute. at each temperature.

4.6.4 Vibration test. The cylinder shall be subjected to a vibration test in accordance with MIL-STD-167-1 to determine conformance with 3,3.5. The cylinder shall be pressurized with gas to room temperature operating pressure and mounted with brackets to simulate installation.

4.6.5 Thermal shock test. Cylinder shall be pneumatically pressurized to operating pressure at room temperature and sealed and soake at minus 20°F for a period of 6 hours. Cylinder shall be transferred to an atmosphere at 160°F and soaked for a period of not less than 2 hours. Throughout this test, the cylinder shall remain sealed with pressure allowed to fluctuate with temperature. The transfer of the cylinder from one temperature chamber to another shall be made in no more than 5 minutes. The cylinder shall meet the requirements of 3.3.4.

4.6.6 Endurance test. Two cylinders shall be pressurized to 3700 lb/in^2 at a temperature of minus 20°F, for not less than 72 hours, and then pressurized to 6300 lb/in^2 and soaked at a temperature of 160°F for not less than 72 hours. They then shall be subjected to cyclic pressure application at a rate not to exceed 10 cycles per ‘minute (c/m) as follows:

(a) Apply 100 cycles (O lb/in^2 to 5000 lb/in^2 to O lb/in^2) at room temperature.

(b) Apply 3500 cycles at pressure excursions of 3700 lb/in^2 to 6300 lb/in^2. Test shall be conducted in an atmosphere of 120°F and not less than 95 percent relative humidity. Upon completion of this exposure and within 24 hours of removal from humidity, one cylinder shall be sujected to the pneumatic leakage test (see 4.6.3.1) and to the hydrostatic burst test as specified in 4.6.7. The endurance test of the other cylinder shall then be continued by an additional 6500 pressure excursions in the

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humidity environment. The cyclic rate shall not exceed 10 c/m. Within 24 hours of removal from the humid environment, the cylinder shall then be subjected to the hydrostatic burst test (see 4.6.7) to determine conformance with 3.3.3. For delay in burst testing see 4.6.9.

4.6.7 Hydrostatic burst test. The cylinder shall be hydrostatically pressurized for 3 minutes to the specified minimum burst pressure at a temperature of from 65°F to 95°F. After the 3 minute period, the pressure shall then be increased to failure of the cylinder. The burst pressure and type of failure shall be recorded. The rate of pressure rise shall be in accordance with 4.5.3.

4.6.8 Fragmentation test. Fragmentation test shall be conducted as specified in 4.6. 8.1 through 4.6.8.3.

4.6.8.1 Pressure. The cylinder shall be pneumatically charged to 5000 lb/in^2 pressure ambient temperature.

4.6.8.2 Fragmentation test conditions. The ammunition shall be 0.30 caliber armor piercing and shall be fired at a muzzle velocity of 2800 + 100 feet per second (ft/s). The range from the gun to the cylinder snail-be not more than 50 yards.

4.6.8.3 Failure criteria. The cylinder shall meet the requirements of 3.3.6.

4.6.9 Soak cycle test. The cylinder shall be placed in a chamber that has an environment of 120°F and not less than 95 percent relative humidity. The cylinder shall be pressurized hydrostatically to 6300 lb/in^2. Each day the cylinder shall be cycled from 6300 lb/in^2 to zero pressure to 6300 lb/in^2 for a total of 4 cycles per day at a rate of approximately 1 elm. Between cycles, the pressure shall be maintained at 6300 lb/in^2. Test shall be continued for at least 30 consecutive days without evidence of damage. Within 24 hours of removal from the humid environment, the cylinder shall be subjected to the hydrostatic burst test as specified in 4.6.7 and shall meet the requirements of 3.3.7. A cylinder may be transported or temporarily stored so long as the exposure to ambient, prior to burst testing, is less than 24 hours.

4.6.10 Pressure storage test. The cylinder shall be pneumatically pressurized to 6300 lb/in^2 and stored in a continuous environmental condition of 120°F and not less than 95 percent relative humidity for 15 days. Within 24 hours of completion of this exposure, the cylinder shall be subjected to the hydrostatic burst test as specified in 4.6.7 and shall comply with the requirements of 3.3.8. For delay in burst testing see 4.6.9.

4.6.11 Humidity cycle test. The cylinder shall be pressurized to 6300 lb/in^2 and placed in an environment of 120°F at not less than 95 percent relative humidity. After conditioning for at least 96 hours, the cylinder shall be subjected to 500 cycles of pressure from 6300 lb/in^2 to O lb/in^2 to 6300 lb/in^2 without failure. The cycle rate shall not exceed 10 elm. Within 24 hours of completion of the test, the cylinder shall be subjected to the hydrostatic burst test (see 4.6.7) to determine conformance with 3.3.9. For delay in burst testing see 4.6.9.

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4.6.12 Resin content determination. Resin content shall be determined in accordance with FED-STD-406, method 7061. Samples for first article inspection shall be taken from four locations on the cylinder: one adjacent to the port fitting, one adjacent to the opposite or blind fitting, and two approximately 180 degrees apart about the midpoint or equator of the cylinder. One sample from a fitting and one sample from the midpoint shall be delaminated into at least two layers to determine the resin content of inner and outer portions of the fiberglass wall. For quality conformance sampling, two samples shall be taken from approximately the same location on the cylinder wall. The location shall be adjacent to an end fitting on one sample unit and from the midpoint of the cylinder on the other sample unit. One of the samples shall be delaminated into at least two layers. Resin content shall be determined for each solid and delaminated wall sample. Moisture-barrier coating shall be removed prior to testing. Resin content shall be reported for all samples. Additionally, for samples that are delaminated, the average resin content for the total wall thickness (fiberglass portion) shall be reported. It shall be calculated by averaging the values for the delaminated samples, weighing the values according to the proportion of the wall thickness each represents, or by combining the resin content for the total wall thickness. Minimum and average resin contents shall conform to 3.2.4.

4.6,13 Void content. Void content determinations shall be made on full-thickness (undelaminated) samples used for resin content tests (see 4.6.12). Tests shall be performed in accordance with the “percent voids" test of MIL-P-17549. In the event of nonconformance of any one determination, additional samples may be taken adjacent to that sample and the results averaged. Void content shall meet requirements of 3.2.5.

4.6.14 Drop test. The cylinder shall be pressurized to 5000 lb/in^2 and dropped a vertical distance of not less than 10 feet onto a rigid steel plate twice in each of the three following configurations:

(a) Impact on valve end of cylinder.
(b) Impact on end opposite valve.
(c) Impact on side of cylinder.

Subsequent to the drops, the cylinder shall meet the examination (see 4.6.2), extreme temperature leakage (see 4.6.3.2), thermal shock (see 4.6.5) and pressure storage (see 4.6.10) tests.

4.6.15 Pressure cycle test. The cylinder shall be pneumatically pressurized to 5000 lb/in^2. The cylinder shall be subjected to cyclic pressure application at a rate not to exceed 10 c/m, from 5000 lb/in^2 to 0, for 25 cycles.

4.7 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging , packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

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5. PACKAGING

(The preparation for delivery requirements specified herein applies only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2.1).

5.1.1 General preparation for packaging. Each cylinder shall be cleaned end dried. Internal drying shall be accomplished by circulation of heated air until the interior of the cylinder is thoroughly dried. After drying, a corrosion inhibitor conforming to MIL-I-22110, type II shall be sprayed inside the cylinder in accordance with the manufacturer's instructions for the required quantity for the particular volume and surface area. All ports shall then be plugged or capped with oil and moisture resistant material to protect bosses and threads from damage to prevent entrance of air or foreign matter into the cylinder.

5.1.2 Level A. Cylinders shall be individually preserved-packaged to meet the requirements of MIL-P-116, method III and as follows: Each cylinder shall be unit packaged in a fiberboard box conforming to PPP-B-636, class domestic, with type and style at the option of the contractor. Box closure, shall be in accordance with method I of the appendix to the box specification. Inner packing forms, of the same material as the box, shall be provided to anchor and cushion the box contents and to provide a snug fitting, nonshifting load.

5.1.3 Level C. Each cylinder shall be packaged in a manner which will afford 'ion against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. The contractor[s] preservation-packaging procedures may be used when they meet the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Cylinders preserved-packaged as specified (see 5.1) shall be packed for shipment in any of the following containers at the contractor's option.

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2

Shipping containers shall be closed, strapped, or banded in accordance with the applicable container specification or appendix thereto. Unless otherwise specified (see 6.2.1), shipping containers shall have case liners conforming to MIL-L-10547. Shipping containers shall not exceed the gross weight limitations of the applicable box specification. Wood , plywood , or wood-cleated boxes exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with the box specification.

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5.2.2 Level B. Cylinders preserved-packaged as specified (see 5.1) shall be packed for shipment in any of the following containers at the contractor's option.

Specification	Type or class
PPP-B-636	Class-domestic
PPP-B-640	Class-domestic
PPP-B-601	Domestic
PPP-B-621	Class 1

Shipping containers shall be closed in accordance with the applicable box specification or appendix thereto, with method I closure applicable to PPP-B-636 boxes. Shipping containers shall not exceed the gross weight limitations of the applicable box specification. Wood , plywood, or wood-cleated boxes exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with the box specification.

5.2.3 Level C. Cylinders preserved-packaged as specified shall be packed for shipment in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing and method of shipment shall comply with Uniform Freight or National Motor Freight Classification rules and regulations or other carrier rules as applicable to the mode of transportation.

5.2.4 Palletization. Alternatively to level B packing, cylinders packaged as specified in 5. 1 may be palletized In accordance with MIL-STD-147.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2.1), interior packages, exterior shipping containers, and palletized loads shall be marked in accordance with MIL-STD-129 and MIL-P-116.

6. NOTES

6.1 Intended use. Cylinder covered by this specification are intended for use in lifeboat inflation equipment conforming to MIL-I-24490, for storage of air at high pressure.

6.1.1 Type I. These cylinders are for use with the MK 5, Mod 2 lifeboat for 15 persons.

6.1.2 Type II. These cylinders are for use with MK 6 lifeboat for 25 persons.

6.2 Ordering data.

6,2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) First article inspection item markings (see 4.3.1).

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(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) When case liners are not required (see 5.2.1).
(f) Special marking required (see 5,3).

6.2.2 Data requirements. When this specification is used in a contract which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
4.3.2.1	First article inspection report	DI-T-4902	----- .

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquis ition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish-evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 1940-N048)

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MIL-C-24604 (SH)

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CONTRACT NUMBER

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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